Introduction to Algorithms

A Sample Dummy Textbook

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Chapter 1: The Role of Algorithms in Computing

An algorithm is any well-defined computational procedure that takes some value, or set of values, as input and produces some value, or set of values, as output. An algorithm is thus a sequence of computational steps that transform the input into the output.

We can also view an algorithm as a tool for solving a well-specified computational problem. The statement of the problem specifies in general terms the desired input/output relationship, and the algorithm describes a specific computational procedure for achieving that relationship.

Algorithms are widely used throughout modern life. Consider a bank that needs to process financial transactions, a search engine that needs to rank web pages, or a navigation app that must find the shortest route between two locations. Each of these tasks relies on algorithms operating efficiently, even as the size of the input data grows very large.

This chapter introduces the basic terminology used throughout the rest of this text, including notions of correctness, efficiency, and the trade-offs designers must consider when choosing between competing approaches to the same problem.

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